



NEWSLETTER

NOVEMBER 2024

AXLES AND FATIGUE

WILL OUR AXLES
SURVIVE TESTING AND
THE DYNAMIC EVENTS?

VISION 2026

LEARN ABOUT WHAT WE
HAVE PLANNED MOVING
FORWARD!



WHAT CAN YOU EXPECT?

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WHAT IS AUS RACING

AUS Racing is the Formula Student team from the American University of Sharjah. Formula Student is an international engineering design competition that graciously provides a platform to aspiring engineers to build an open-wheel, single seater racecar and simulate all the different nuances that culminate to account for its structural integrity, performance, economic feasibility and industrial standards.

Students compete in different static and dynamic events to assess the car, thus testing the engineering design, management skills and the overall team spirit of the contesting teams. It is an esteemed competition that unlocks new horizons for AUSers

to venture into as we progress into a new era of technological competence and innovation.

Here at AUS Racing, we hope to open up opportunities for students for the foreseeable future to take part in a project that will teach them engineering, management and budgeting. In essence, real world engineering.

TORQUING THE STEERING



Why does a car drive straight and tend to follow its direction of motion, even when you are not touching the steering wheel? The answer lies in the self aligning torque.

Self aligning torque is the tendency of the wheels to align themselves with the direction of motion using caster.

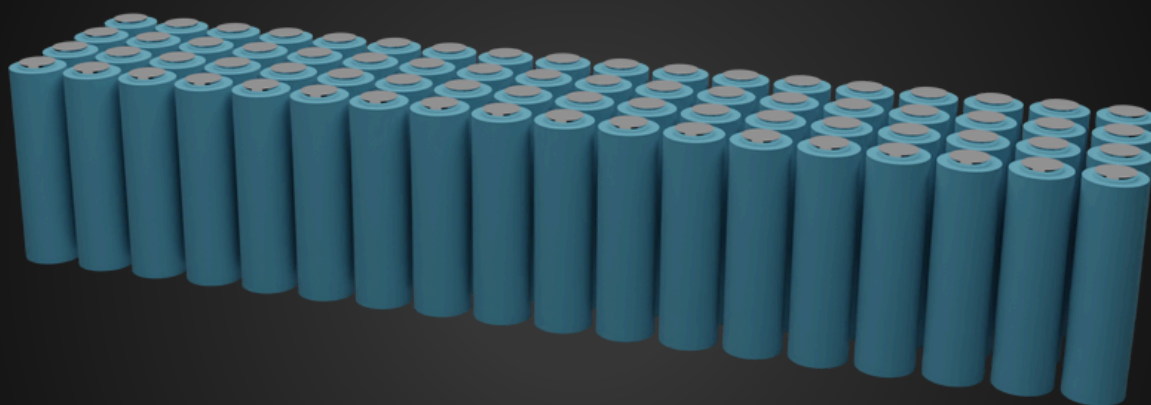
Caster uses the same principle as the wheels on a trolley or a

shopping cart. More caster is better as it makes the car more stable and easier to drive.

However, too much self aligning torque will make the steering wheel itself difficult to turn.

In order to decide how much torque was too much, we designed a simple rig for our drivers to practice on. With these numbers we can get the most performance out of a drivable car.

ORGANISING THE CELLS



Selecting an electrical configuration for the cells is simple. The only consideration is getting the required energy and voltage requirements.

The physical configuration on the other hand needs to be modular, volumetrically efficient and capable of sustaining lateral, longitudinal and horizontal Gs.

Modularity is extremely important for designing and troubleshooting. Dividing the battery into accessible segments allows ease of access to the problematic part of the battery if the need arises.

With the modularity, it also needs to be volumetrically efficient as each segment needs to be insulated from the next and fire resistant for safety.

LONGEVITY OF OUR AXLES



Designing the driveshaft and other axles is not as simple as making it thick enough to withstand the torsional loads.

The main considerations while designing axles involves reducing the weight of the axles as well as longevity.

Since the axles need to be able to run at 1000 RPM, they undergo a

vast number of cycles throughout their lifetime.

Furthermore, they also undergo bending loads at the bearings on the uprights.

All of these loadings need to be accounted for while working with the fatigue or longevity characteristics of the axle.

THE 2026 VISION

The American University of Sharjah's first Formula Student team is set to debut in the Concept Class at FSUK 2025, marking a significant milestone for the university. The team's innovative approach and relentless dedication have positioned them as pioneers in the Middle East's Formula Student landscape.

Building on this foundation, they aim to compete with a fully functional EV model in FSUK 2026, showcasing their commitment to sustainability and cutting-edge engineering. This two-year journey highlights the team's progressive vision, combining academic rigor with practical expertise to excel on the global stage and redefine the future of motorsport innovation.



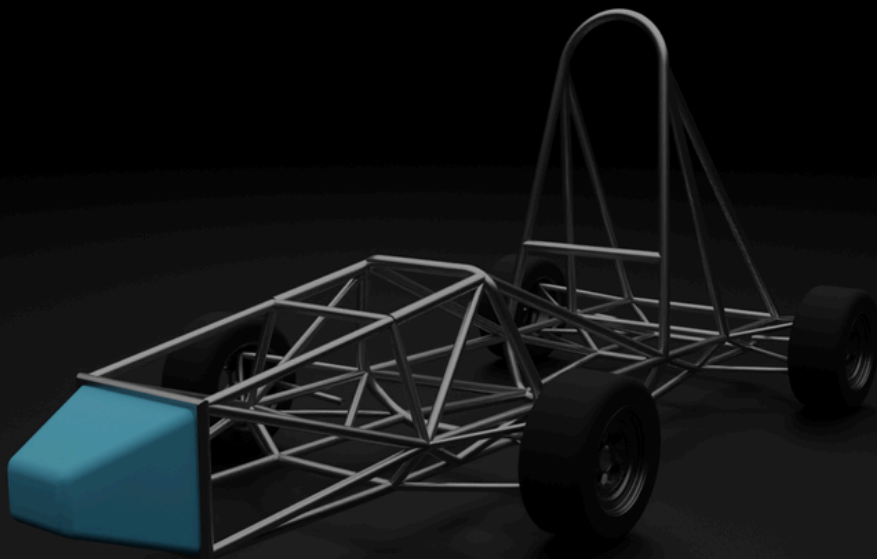
HOW YOU CAN SUPPORT US

What if you had the front row seat to engineering pioneers at work?

What if you were the reason these budding innovators were changing the future?

What if your institution could get their message to the rest of the world through Formula Student?

If your company's answer to any of these questions is yes, contact aaaa@aus.edu to learn more about getting involved with the AUS Racing team.



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American
University
of Sharjah

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